Tumhare course ke concepts jo Data Mining ke paper me aa sakte hain:

Python/ML Course	Data Mining Paper
CSV Files	Data Sources
NumPy	Data Handling
Pandas	Data Preprocessing
Scatter Plot	Data Visualization
K-Means	Clustering
Supervised Learning	Classification
Unsupervised Learning	Clustering
Train-Test Split	Model Evaluation
Confusion Matrix	Performance Evaluation
Feature Scaling	Data Preprocessing
Classification	Data Mining Task
Regression	Prediction Task

K-Means

Tum ne jo K-Means padha tha:

- Unsupervised Learning
- Data ko groups me divide karta hai
- Har group ka center = Centroid

Ye **Data Mining ka bohat important topic** hai.

Confusion Matrix

Tum ne jo T+, T-, F+, F- wale concepts puche thay:

Actual	Predicted	Name
Yes	Yes	True Positive
No	No	True Negative
No	Yes	False Positive
Yes	No	False Negative

1. AI → ML → Data Mining → Data Science

Artificial Intelligence (AI)

Machine ko intelligent banana.

Example:

- ChatGPT
- Self-driving car
- Face recognition

Machine Learning (ML)

AI ka subset hai.

Machine data se khud seekhti hai.

Example:

- Email spam detection
- House price prediction

Data Mining

Large data se hidden patterns aur knowledge nikalna.

Example:

- Supermarket customers ki buying patterns

Data Science

Data collect + clean + analyze + visualize + ML apply.

Example:
Business reports banana.

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2. Data Mining Process

Steps:

1. Data Collection
2. Data Cleaning
3. Data Integration
4. Data Selection
5. Data Transformation
6. Data Mining
7. Pattern Evaluation
8. Knowledge Presentation

Example

Store ka data:

Customer	Product
A	Milk
B	Bread

Data clean karo → analyze karo → pattern nikalo.

---

3. KDD Process

KDD = Knowledge Discovery in Databases

Data se knowledge discover karna.

Steps

1. Selection
2. Cleaning
3. Transformation
4. Data Mining
5. Evaluation
6. Knowledge

Difference

KDD

- Complete process

Data Mining

- KDD ka sirf ek step

Exam Question:
Difference between KDD and Data Mining?

Answer:
KDD complete process hai jabke Data Mining KDD ka ek stage hai jahan patterns discover kiye jate hain.

---

4. Classification

Supervised Learning technique.

Data ko predefined classes me divide karta hai.

Example

Marks	Result
80	Pass
20	Fail

Model seekhta hai aur new student ko Pass/Fail batata hai.

Examples

- Spam / Not Spam
- Pass / Fail
- Disease / No Disease

---

5. Clustering

Unsupervised Learning technique.

Data ko similar groups me divide karta hai.

Example

Customers ko groups me divide karna:

- Students
- Businessmen
- Professionals

Important

K-Means = Most famous clustering algorithm

Centroid

Cluster ka center point.

Classification vs Clustering

Classification	Clustering
Supervised	Unsupervised
Labels hote hain	Labels nahi hote
Pass/Fail	Similar groups
Known classes	Unknown classes

---

6. Association Rules

Items ke beech relationship discover karna.

Example

Supermarket:

Milk khareedne wale Bread bhi khareedte hain.

Rule:

Milk → Bread

Uses

- Market Basket Analysis
- Product Recommendation

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7. Apriori Algorithm

Association Rules find karne ke liye use hota hai.

Purpose

Frequently purchased items dhoondhna.

Example

Transactions:

T1 = Milk, Bread

T2 = Milk, Butter

T3 = Milk, Bread

Apriori dekhega:

Milk sabse zyada aa raha hai.

Rule:

Milk → Bread

---

8. Support and Confidence

Support

Kitni transactions me item aaya.

Support(Milk)

= Milk wali transactions / Total transactions

Confidence

Rule kitni dafa correct nikla.

Confidence(Milk → Bread)

= Milk aur Bread dono / Milk

---

9. Data Warehouse vs Data Mining

Data Warehouse	Data Mining
Data store karta hai	Knowledge nikalta hai
Huge database	Analysis technique
Historical data	Patterns discover
Storage	Intelligence

Example

Warehouse = Library

Data Mining = Library me useful book dhoondhna

---

10. Supervised Learning

Input + Output dono diye jate hain.

Examples

- Classification
- Regression

Example

Study Hours	Marks
2	40
4	70

Machine relation seekh leti hai.

---

11. Unsupervised Learning

Output labels nahi diye jate.

Machine khud patterns find karti hai.

Example

- Clustering
- K-Means

---

12. Regression

Continuous value predict karta hai.

Example

House Price Prediction

Area	Price
5 Marla	40 Lakh

Output number hota hai.

13. Train-Test Split

Dataset ko do parts me divide karte hain.

Train Data

Model seekhta hai.

Test Data

Model ko check karte hain.

Mostly:

- 80% Train
- 20% Test

---

14. Confusion Matrix

Actual / Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

TP

Pass ko Pass predict kiya.

TN

Fail ko Fail predict kiya.

FP

Fail tha lekin Pass predict kar diya.

FN

Pass tha lekin Fail predict kar diya.

---

15. Accuracy

Model kitni baar sahi nikla.

Accuracy = Correct Predictions / Total Predictions

---

16. F1 Score

Precision aur Recall ka combination.

Jab classes imbalance hon tab useful.

---

17. Feature Scaling

Values ko same scale par lana.

Example

Age = 20

Salary = 100000

Salary bohat badi value hai.

Scaling ke baad dono comparable ho jate hain.

---

18. NumPy

Numerical calculations ke liye.

Uses

- Arrays
- Mathematical operations
- Fast computation

---

19. Pandas

Data handling ke liye.

Uses

- CSV files
- Tables
- Data analysis

---

20. CSV File

Comma Separated Values

Example:

```
Name,Age
Ali,20
Ahmed,22
```

---

21. Scatter Plot

Data points show karta hai.

Axes

X-axis = Horizontal

Y-axis = Vertical

Har dot ek observation hota hai.

---

22. K-Means (Very Important)

1. K select karo

2. Centroids choose karo
3. Points assign karo
4. Centroids update karo
5. Repeat until stable

Output

Clusters

Example

Students ko performance groups me divide karna.

---

5 Most Important Exam Definitions

Data Mining

Large datasets se hidden patterns discover karna.

KDD

Database se useful knowledge discover karne ka complete process.

Classification

Predefined classes me data assign karna.

Clustering

Similar objects ko groups me divide karna.

Apriori Algorithm

Frequent itemsets aur association rules discover karne wala algorithm.

1. Model Train Karna

Train karna = Machine ko data se sikhana

Example:

Study Hours	Result
2	Fail
4	Pass
6	Pass

Machine ye relation seekh leti hai.

Phir new data do:

Study Hours = 5

Machine predict karegi: **Pass**

Isi learning process ko **Training** kehte hain.

---

2. Actual vs Predicted

Actual

Real answer

Predicted

Machine ka answer

Example:

Student	Actual	Predicted
A	Pass	Pass

Student	Actual	Predicted
B	Fail	Pass

Actual = Reality

Predicted = Model ka guess

---

3. Types of Machine Learning

Supervised Learning

Input + Output dono diye jate hain.

Example:

Hours	Marks
2	40
4	80

Algorithms:

- Classification
- Regression

Unsupervised Learning

Sirf input hota hai.

Machine khud patterns dhoondti hai.

Algorithms:

- K-Means
- Clustering

Reinforcement Learning

Reward aur punishment se seekhta hai.

Example:

- Game playing AI
- Robots

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4. Classification

Categories predict karta hai.

Output fixed classes hoti hain.

Example:

- Pass / Fail
- Spam / Not Spam
- Male / Female

Output = Label

---

5. Regression

Number predict karta hai.

Example:

Area	Price
5 Marla	40 lakh

Output = Continuous Value

Examples:

- House Price
- Temperature
- Salary

Classification vs Regression

Classification	Regression
Categories	Numbers
Pass/Fail	House Price
Spam/Not Spam	Temperature

---

6. Clustering

Similar objects ko groups me divide karta hai.

Output labels pehle se nahi hote.

Example:

Students ko groups me divide karna:

- Good
- Average
- Weak

Machine khud groups banati hai.

---

7. K-Means Clustering

Most famous clustering algorithm.

K Means

K = Number of clusters

Example:

K = 3

To 3 groups banenge.

Steps

1. K choose karo
2. Centroids select karo
3. Points assign karo
4. New centroid calculate karo
5. Repeat

Centroid

Cluster ka center point.

---

8. Accuracy

Kitni predictions sahi hui.

Formula:

Accuracy=Correct PredictionsTotal PredictionsAccuracy=\frac{Correct\ Predictions}{Total\ Predictions}Accuracy=Total PredictionsCorrect Predictions

Example:

100 predictions me 90 sahi

Accuracy = 90%

---

9. Confusion Matrix

Actual/Predicted	Positive	Negative
Positive	TP	FN

Actual/Predicted	Positive	Negative
Negative	FP	TN

TP

Pass tha → Pass predict

TN

Fail tha → Fail predict

FP

Fail tha → Pass predict

FN

Pass tha → Fail predict

---

10. Precision

Jitne Positive predict kiye un me kitne sahi the.

Example:

10 students ko Pass bola

8 waqai Pass the

Precision = 8/10

---

11. Recall

Jitne asal me Positive the un me kitne pakre gaye.

Example:

Actual Pass = 20

Model ne 15 identify kiye

Recall = 15/20

---

12. F1 Score

Precision aur Recall ka balance.

Formula:

F1=2×Precision×RecallPrecision+RecallF1=2\times\frac{Precision\times Recall}{Precision+Recall}F1=2×Precision+RecallPrecision×Recall

Exam me yaad rakho:

F1 Score = Precision aur Recall ka harmonic mean

---

13. Entropy

Decision Tree me use hoti hai.

Data kitna mixed ya impure hai.

Example

Class me:

50 Pass

50 Fail

Bohat confusion hai.

Entropy High

Example 2

100 Pass

0 Fail

Koi confusion nahi.

Entropy Low

Yaad rakho:

More Disorder = High Entropy

Less Disorder = Low Entropy

---

14. CSV

CSV = Comma Separated Values

Example:

```
Name,Age,Marks
Ali,20,80
Ahmed,21,70
```

Python me:

```
import pandas as pd

df = pd.read_csv("file.csv")
```

---

15. NumPy

Numerical calculations ke liye.

Example:

```
import numpy as np

a = np.array([1,2,3])
```

Uses:

- Arrays
- Mathematics
- ML calculations

---

16. Pandas

Data handling ke liye.

Example:

```
import pandas as pd
df = pd.read_csv("students.csv")
```

Uses:

- CSV files
- Tables
- Data analysis

Sir ke Quiz/Paper ke Liye 1-Line Ratta Sheet

☐ Data Mining = Hidden patterns discover karna

☐ Model Training = Machine ko data se sikhana

☐ Classification = Categories predict karna

☐ Regression = Numbers predict karna

☐ Clustering = Similar groups banana

☐ K-Means = Most popular clustering algorithm

☐ Centroid = Cluster ka center

☐ Actual = Real answer

☐ Predicted = Model ka answer

- Accuracy = Overall correctness
- Precision = Predicted positives me sahi positives
- Recall = Actual positives me pakre gaye positives
- F1 Score = Precision + Recall ka balance
- Entropy = Disorder/Impurity measure
- CSV = Comma Separated Values file
- Supervised = Labels available
- Unsupervised = Labels unavailable

Data Preprocessing

Data ko machine learning ya data mining ke liye tayyar karna.

Real-world data aksar:

- Incomplete hota hai
- Noisy hota hai
- Duplicate hota hai
- Missing values hoti hain

Is liye pehle preprocessing karte hain.

Steps of Data Preprocessing

1. Data Cleaning

Errors aur missing values remove karna.

Example:

Name	Age
Ali	20

Name	Age
Ahmed	NULL

NULL value ko fill ya remove karenge.

---

2. Data Integration

Different sources ka data combine karna.

Example:

- Student Database
- Fee Database

Dono ko merge karna.

---

3. Data Transformation

Data ko suitable format me convert karna.

Example:

Age:

20, 25, 30

ko scale kar dena.

---

4. Data Reduction

Large data ko chhota karna.

Purpose:

- Fast processing
- Less storage

---

5. Feature Scaling

Values ko same scale pe lana.

Example:

Age	Salary
20	100000

Salary ki value bohat bari hai.

Scaling ke baad dono comparable ho jate hain.

Why Data Preprocessing Important?

Without preprocessing:

 Low Accuracy

 Bad Predictions

 Slow Models

With preprocessing:

 Better Accuracy

 Better Performance

 Better Predictions

---

2. SciPy Library

SciPy = Scientific Python

NumPy ke upar bani hui library hai.

Scientific aur mathematical calculations ke liye use hoti hai.

Import:

```
import scipy
```

SciPy Uses

Mathematics

Complex calculations

Example:

- Integration
- Differentiation
- Equations

Statistics

Mean, probability, distributions

Example:

```
from scipy import stats
```

Optimization

Best solution find karna.

Example:

Minimum cost

Maximum profit

Linear Algebra

Matrices aur vectors

Example:

```
from scipy import linalg
```

Signal Processing

Audio aur signals process karna.

Image Processing

Images analyze karna.

NumPy vs SciPy

NumPy	SciPy
Arrays	Advanced Scientific Computing
Basic Mathematics	Advanced Mathematics
Foundation	Built on NumPy
Faster Arrays	Specialized Algorithms

Exam Short Definitions

Data Preprocessing

Raw data ko clean aur transform karke analysis ke liye prepare karne ka process.

Data Cleaning

Missing, noisy aur incorrect data remove karna.

Feature Scaling

Features ko same range me convert karna.

SciPy

Scientific and technical computing ke liye Python library.

NumPy

Numerical computations aur arrays ke liye Python library.

Quick Revision (1 Marks)

Data Preprocessing?
→ Raw data ko ML/Data Mining ke liye prepare karna.

Data Cleaning?
→ Missing aur incorrect values remove karna.

Feature Scaling?
→ Features ko same scale par lana.

SciPy?
→ Scientific computing library.

NumPy?
→ Numerical arrays library.

Pandas?
→ Data handling aur CSV files ke liye.

Ye 3 libraries yaad rakho:

- **NumPy** → Arrays & Math
- **Pandas** → Tables & CSV
- **SciPy** → Scientific & Statistical Calculations